TYLER FRANKE

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EDUCATION

Universität Hamburg *Dec. 2024 - Dec. 2027 (expected)*
Dr. rer. nat.
Project A4 within CRC 1624 "Higher structures, moduli spaces and integrability"

Thesis title: Integrable systems and defect skein theory
Supervised by Prof. Dr. Ingo Runkel

University of Melbourne *Mar. 2021 - Dec. 2023*
Master of Science (Mathematics and Statistics) (with Distinction)
Mathematical Physics and Physical Combinatorics specialisation

Thesis title: String order in the q-deformed Affleck-Kennedy-Lieb-Tasaki model
Supervised by Dr. Thomas Quella

University of Melbourne *Mar. 2017 - Dec. 2019*
Bachelor of Science
Mathematical Physics major

PUBLICATIONS

· **T. Franke**, T. Quella, *Duality and hidden symmetry breaking in the q-deformed Affleck-Kennedy-Lieb-Tasaki model*, doi: 10.21468/SciPostPhysCore.7.4.078.

IN-PREPARATION

· *Braided module categories from small braidings*, joint with Ingo Runkel, Volker Schomerus. Draft available here.

· *Skein modules with line and point defects*, joint with Ingo Runkel, Volker Schomerus.

INVITED TALKS

· *Braided module categories from small braidings*, Séminaire de Topologie, Institut de Mathématiques de Jussieu-Paris Rive Gauche (Paris, France), October 2026.

· *Constructions of braided module categories*, Algebra and mathematical physics research seminar, Universität Hamburg (Hamburg, Germany), May 2026.

· *Classifying 2d gapped phases with defects via the SymTFT*, Universität Hamburg, SymTFT mini course, March 2025.

· *Dualities in the q-deformed Affleck-Kennedy-Lieb-Tasaki model*, University of Melbourne Mathematical Physics seminar, University of Melbourne (Melbourne, Australia), October 2024.

CONTRIBUTED TALKS

Integrable systems and skein theory, Summer School CRC 1624: Higher structures, moduli spaces and integrability, gong-show presentation, Universität Hamburg (Hamburg, Germany), August 2026.

· *Turaev–Viro TQFT's and the relation with the WRT TQFT*, BTW6: The Volume Conjecture: From Some TQFTs to Recent Developments (Fréjus, France), May 2026.

· *The reflection equation*, Junior ZMP Seminar, Deutsches Elektronen-Synchrotron (Hamburg, Germany), January 2026.

· *Integrable systems and skein theory*, The Physics and Mathematics of Boundaries, Impurities, and Defects, gong-show presentation, Isaac Newton Institute (Cambridge, United Kingdom), September 2025.

· *Integrable systems and skein theory*, Higher Structures: Recent developments and applications (Summer school), gong-show presentation, Universität Hamburg (Hamburg, Germany), September 2025.

· *Dualities in the q-deformed Affleck-Kennedy-Lieb-Tasaki model*, Tensor Categories, Quantum Symmetries, and Mathematical Physics, contributed talk, Matrix Institute (Creswick, Australia), November 2024.

· *Duality and hidden symmetry breaking in the q-deformed Affleck-Kennedy-Lieb-Tasaki model*, Symmetries 2024, poster and gong-show presentation, University of Oxford (Oxford, United Kingdom), August 2024.

· *Dualities in the qAKLT model*, Mathematical physics semester kick-off, University of Melbourne, February 2024.

· *Introduction to fusion categories*, University of Melbourne, Categorical symmetries seminar, August 2022.

SERVICE

· *Postgraduate Programs Committee*, Postgraduate coursework representative, University of Melbourne, *Mar. 2023 - December 2023*

· *Statistics and Mathematics Postgraduate Student Society*, University of Melbourne

  · Communications and publicity officer: *May 2023 - June 2024*

  · General committee member: *May 2022 - May 2023*

TEACHING

· *Introduction to category theory*, bachelor seminar coordinator, Universität Hamburg, Winter Semester '26/'27.

· *Methods of mathematical physics*, teaching assistant, University of Melbourne, '23 and '24.

· *Linear algebra*, teaching assistant, University of Melbourne, '24.

· *Calculus 2*, teaching assistant, University of Melbourne, '23.

HONORS AND AWARDS

· University of Melbourne, Melbourne, Australian Government Research Training Program Scholarship (Stipend) - 2024

· University of Melbourne, Melbourne, Mathematics and Statistics Masters Scholarships - 2021 & 2022